When should a lunar lander believe its altimeter?

A controlled comparison of sensor-trust rules on real South Pole terrain (draft, 19 October 2026)

Abstract

Two recent lunar landers were lost partly because their navigation filters misjudged a sensor: a valid altimeter reading at a crater edge was rejected as a fault, and a camera was trusted in shadow. We build a seeded descent simulator on NASA PGDA Site04 (Shackleton rim), measure a landmark-matching camera's reliability in a Unity renderer instead of assuming it, and compare seven ways of judging a surprising altimeter reading over 500 landings per scenario, with a pre-registered design and a held-out site. A fixed-gate filter with a coarse reference reproduces the HAKUTO-R failure (0.2% landing success). Putting the terrain map in the filter recovers most of it (82%), and a plain map-tolerance gate reaches 97%. A four-step trust rule with probation matches but does not beat that gate in cliff scenarios, wins with a noisy altimeter in shadow (88% vs 64%) and with no map (91% vs 3%), and fails when the altimeter is noisy in good light (39% vs 99%) or the map is shifted (1% vs 35%). Camera fixes must be gated by the matcher's own confidence (ungated: under 3% success); predicting darkness from Sun and terrain adds nothing. Delayed readings are solved by time stamps, not by trust logic. We report limits and the points where an earlier, unfairly designed comparison had overstated the benefit.

1. Introduction

Powered descent navigation fuses a laser altimeter, a camera and an IMU. A fixed chi-square gate cannot tell a real terrain discontinuity from a sensor fault. HAKUTO-R M1 (April 2023) rejected altimeter readings after crossing a crater rim; IM-2 (March 2025) kept trusting a camera that could not see in shadow. PITFAL-style filters put a terrain map into the prediction (Adams et al., 2025). We ask which additional judging rule, if any, is worth having once the map is in the filter and the camera is handled honestly.

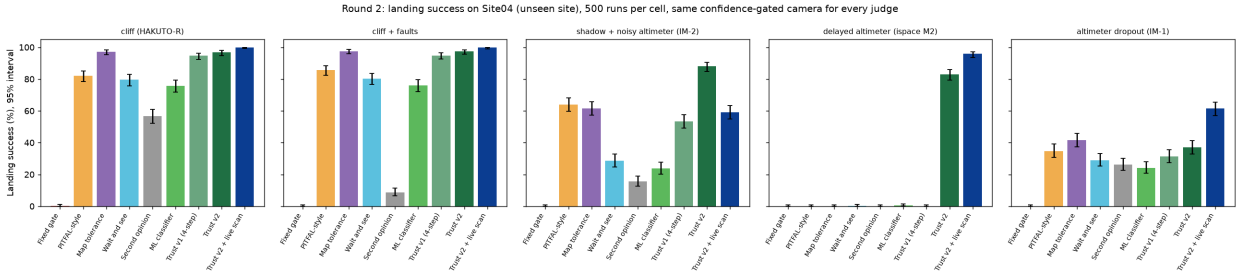
2. Method

Terrain. PGDA Product 78 (Barker et al., 2021): Site04 (unseen test) and Site01 (tuning, with NASA's total-error map) at 5 m/pixel, resampled to 10 m; procedural crater worlds for tuning and classifier training. The simulated Moon contains unmapped craters and roughness absent from the onboard map, plus a registration shift (fair-testing rule 1). **Vehicle and guidance.** Point-mass lander, lunar gravity, throttleable engine with fuel; guidance flies a speed profile on the estimated terrain-relative altitude. Success: vertical speed <= 2 m/s, horizontal <= 1.5 m/s, slope <= 10 degrees, fuel left. **Sensors.** IMU with bias and noise; laser altimeter with range-dependent noise and fault bursts; camera position fixes drawn from a Unity survey (300 image pairs, normalised cross-correlation matcher, direct-light-only prediction); optional live lidar scan. **Judges.** Fixed gate with coarse reference; PITFAL-style (map in prediction, noise inflation after a flag); map tolerance (gate widened by local terrain variance and map error); wait-and-see (probation); second opinion; random-forest classifier (trained on procedural worlds only); and the four-step trust rule (physical possibility, probation, camera, no-map fallback) in two versions. **Scenarios.** Cliff crossing, cliff with frequent faults, shadow with a noisy altimeter, delayed readings, altimeter dropout. All judges see identical noise and faults. **Protocol.** Hypotheses and settings were fixed in a pre-registration committed before the Site04 run; tuning used procedural worlds and Site01 only; 500 runs per cell with 95% intervals; an exploratory control was added afterwards and is labelled.

3. Results

Scenario	Fixed	PITFAL	Map tol.	Trust v1	Trust v2
Cliff (HAKUTO-R)	0.2	82.0	97.2	94.6	96.8
Cliff + faults	0.0	85.6	97.6	94.8	97.4
Shadow + noisy altimeter	0.0	64.0	61.6	53.4	88.0
Delayed readings (v2 time-stamped)	0.0	0.0	0.0	0.0	83.0
Altimeter dropout	0.0	34.8	41.6	31.4	37.0

Table 1. Landing success (%), Site04, 500 runs per cell, confidence-gated camera for every judge.



Camera. The matcher is accurate (median 0.8-2.4 m) once a quarter of the footprint is lit and fails in the dark (88% of fixes off by more than 100 m); its confidence flags every failure in our survey (0 of 226 accepted fixes off by >100 m). Using every fix: 0.4-3% success. Sun-geometry prediction instead of confidence: 5-11 points worse in three of four cases. **Delays.** With time stamps PITFAL-style reaches 74%, the tolerance gate 97.6% and trust v2 83%; without them trust v2 reaches 0%. **Robustness.** No map: 91% vs 3%; divert: 91% vs 74%; position uncertainty 400 m: 96% vs 61%; unseen Site04 98% vs 82% (Site01: both 100%). Failures: noisy altimeter in good light 39% vs 99.5%; map shifted by 2 km 1% vs 35%. Live scan adds up to 24 points in three scenarios and removes 29 in the shadow + noisy one.

4. Discussion

The map in the filter does most of the work; a plain tolerance gate does the rest in cliff cases. The four-step rule's distinctive parts (physical-possibility check, probation) add little there and introduce brittleness: the check rejects valid noisy readings when position is well known and any reading when the map is wrong. Its learned noise level is what helps with a noisy altimeter, at the cost of accepting more faulty readings. For camera handling the practical lesson is to trust the matcher's confidence. An earlier version of this study compared a shadow-aware camera rule with judges that used every fix, and reported a large benefit; with a measured camera and a fair baseline that benefit disappeared.

5. Limitations

- One simulator, one renderer and matcher, one unseen real site; no hardware, no real image-based navigation, no real HAKUTO-R altitude record.
- Settings were first explored while looking at Site04 runs; the pre-registration limits this but does not erase it (disclosed in the registration).
- Trust v2 assumes time-stamped readings and a registered map.
- The no-map baseline uses a coarse reference rather than a terrain-relative filter.

6. Conclusion

For this class of descent, a map-tolerance gate with a confidence-gated camera and time-stamped altimeter readings is about as good as the four-step trust rule in the cliff cases. The trust rule is worth keeping only for its noise-learning, no-map fallback and divert behaviour, and it needs a physical check that scales with measurement noise and a way to detect a wrong map.

References

Adams et al. (2025), AIAA SciTech, NTRS 20240014100 (PITFAL). Barker et al. (2021), Planetary and Space Science 203 (LOLA 5 m DEMs). Bertone et al., Planetary Science Journal, doi:10.3847/PSJ/ae5b70. Karlgaard et al. (2025), NTRS 20250010290. ispace HAKUTO-R Mission 1 debriefing (2023). Code, data products, seeds and the pre-registration: github.com/AnakinSkywalker0/lantern-sensor-trust (private).